

Journal of the Operational Research Society

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/tjor20

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To cite this article: A. Baskar & M. Anthony Xavior (2021): A four-point direction search heuristic algorithm applied to facility location on plane, sphere, and ellipsoid surfaces, Journal of the Operational Research Society, DOI: 10.1080/01605682.2021.1984185

To link to this article: https://doi.org/10.1080/01605682.2021.1984185

Published online: 05 Oct 2021.

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JOURNAL OF THE OPERATIONAL RESEARCH SOCIETY
https://doi.org/10.1080/01605682.2021.1984185

ORIGINAL ARTICLE

A four-point direction search heuristic algorithm applied to facility location on plane, sphere, and ellipsoid surfaces

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ABSTRACT

A facility location problem (FLP) refers to the best location for establishing a facility to optimally connect and meet the requirements of the demand points. In a supply chain on the earth's surface, the cost function essentially includes a distance metric to compute the objective function. If the data points are located in a small area, Euclidean Distance (ED) can be considered. When the data points are fairly separated on the earth's surface, ED cannot give a correct estimate and we have to opt for either a spherical model or an ellipsoidal model. This paper proposes a simple four-point direction search algorithm and a model which can be used for locating a facility on a plane, sphere, and ellipsoid model of earth. The main advantage of the proposed algorithm is that it can be used with any distance metric without changing the algorithm.

ARTICLE HISTORY
Received 15 June 2021
Accepted 16 September 2021

KEYWORDS
Facility location; Fermat–Weber problem; four-point direction search algorithm; geodetic coordinates; Haversine formulae; Vincenty's formulae

1. Introduction

Facility location problems (FLPs) are an important area of operations research that deals with optimum placement of facilities to minimise transportation and establishment costs (Turkoglu & Genevois, 2019). The objective is to find the most suitable geographical location to establish the facilities.

ReVelle and Eiselt (2005) list four basic features for a good location model:

- The customers are located at points or routes.
- The facilities that shall be located.
- The geographic space or surface where the customers/facilities are located.
- The metric indicates the distance, cost, or time between the locations and facilities.

The location space may be planar, spherical, or ellipsoidal. FLP generally involves a distance metric which may be Euclidean distance (ED), great circle distance (GCD), ellipsoidal distance (ELD), etc. ED is used for planar models, GCD for spherical, and ELD for ellipsoidal models. Initially, most of the authors considered ED to solve such problems which yield reasonably good results when the data points are closely located on the earth's surface. In its general form, the Fermat–Weber location problem (Weber, 1982) is to find a point in $\mathbb{R}^n$ that minimises the sum of the (weighted) EDs from "m" given points in $\mathbb{R}^n$. The facility may be a public utility, manufacturing plant, distribution point amongst others.

The models proposed by Katz and Cooper (1980) and Drezner and Wesolowsky (1978) are more realistic especially for larger area FLPs since they considered GCD. Katz and Cooper considered ED as well as GCD and proposed slightly modified formulae to compute the location iteratively. They also confirmed that if the data points are widely separated, the difference between ED and GCD becomes significant. Xue (1994) proposed a gradient algorithm for solving the spherical FLP and proved its global convergence. One heuristic algorithm was proposed by Levin and Ben-Israel (2004) to solve large-scale multi-FLPs. Kalczynski et al. (2020) defined the multiple weighted obnoxious facilities location problem and solved it heuristically. They maximised the minimum weighted distance between facilities and a given set of communities. A procedure was proposed by Rodríguez-Chía and Valero-Franco (2013) to solve the classical location median problem by measuring the distances with p-norms ($p > 2$). A location-driven method was used by Gao (2020) for locating a warehouse to stock medical supplies. The advantages of the proposed method were demonstrated through several simulation studies.

Several authors used the classical Weiszfeld algorithm (Weiszfeld, 1937; Plastria, 2011) to approximately fix the facility (Baskar, 2020, 2021). However, using the Weiszfeld algorithm for points that are fairly separated on the earth's surface may not yield the required accuracy. The GCD models can yield better results for marine applications

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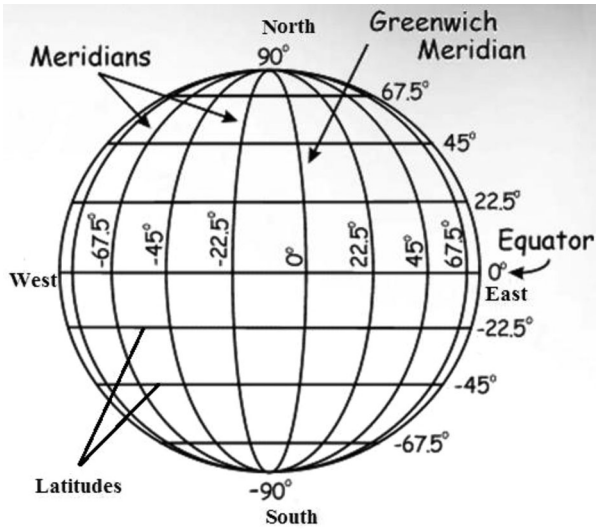


Figure 1. Geodetic coordinates.

(Mwemezi & Huang, 2011; Baskar & Anthony Xavior, 2021) since the data points as well as the final facility locations can be comfortably connected through great circles. Evolutionary algorithms are also widely used by several authors in the facility location domain. An Ant Colony Optimization Algorithm (ACOII) presented recently by Zouein and Kattan (2021) for solving the dynamic FLPs is a good example.

The entire exercise of locating a facility in any space becomes approximate since the actual distance between two points on the earth's surface varies from the computed distance. An approximate factor termed as "Wiggle Factor" is multiplied to make the obtained results more accurate. However, the wiggle factor (curvature of the road) itself is an approximate one that varies from rural to urban, national highways to village roads and of course from country to country. The Wiggle Factor was proposed by Cooper (1983) and computed a value for UK roads. Domínguez-Caamaño et al. (2016), through an improved methodology applied to Spanish road transport, proposed two different Wiggle Factors; one to rural areas and the other one for high-capacity roads. In real cases, the actual or driving distance between two points is more than the ED, GCD, or ELD. The difference becomes significant if constraints like lakes, hills, or rivers are present between the locations. Shih (2015) presented a framework and algorithm to use driving distances on a spherical surface and explores its use as a facility location decision tool. Several exact and heuristic algorithms have been proposed over the years and are available in the literature for the basic vehicle routing problem (VRP) and its variants. Several researchers limit their work to three variants of VRP. The Hybrid Variable Neighborhood algorithm proposed by Penna et al. (2019) was found to handle six variants. Konstantakopoulos et al. (2020) reviewed many such papers published between 2010 and the first quarter of 2020 emphasising the algorithms used for the distribution of goods. The data points may be discrete or continuous. Recently, Zhou et al. (2021) studied a multi-FLPs with demand distributed continuously along a curve (MFLCD) in a planar space instead of discrete points. The computational results show the solution quality of the proposed "alternate location and allocation" (ALA) based algorithm in solving such problems.

It is shown that the true shape of the earth is not a sphere but, an oblate ellipsoid. An ellipsoid is nothing but, a three-dimensional analogue of an ellipse with semi-axes "a" and "b". In an oblate ellipsoid, the third semi-axis "c" = "b" > "a". Hence, the ELD will be more appropriate than GCD (Cazabal-Valencia et al., 2016). The ELD is slightly less than the GCD between two points on the earth. Not many works are available in the literature in this regard. The points on the earth's surface can be conveniently represented by geodetic coordinates (Figure 1); latitudes, longitudes, and altitudes. Most of the models assume that all the data points are located at sea level to avoid any complexity. Hence, altitudes (elevation) are ignored. Latitudes are the imaginary lines that run in an east–west direction along Earth's surface. The equator that divides the Earth's Northern and Southers hemispheres is assumed to be located at 0° latitude (Understanding Maps of Earth, n.d.). The magnitudes of latitude vary between 0° and 90°. They are followed by the letter "N" if they represent the points located north of the equator and by "S" for southern locations.

On the other hand, longitudes or meridians are the imaginary lines that run between the north and south poles. The longitude passing through Greenwich is termed as the prime meridian and has a magnitude of 0°. Unlike the latitudes, the longitudes are specified by both positive and negative values between 0° and 180° followed by the letter E or W. If the angles are to be represented in radians, 180° equals to "π" radians. This paper proposes a simple four-point direction search heuristic algorithm applied to FLP with any distance metric. The model is validated using the datasets available in the literature. This work considers the shortest distance without considering the actual driving distance.

2. Mini-sum and mini–max location problems

FLPs are often formulated in one of two ways, mini-sum problem, and mini–max problem.

2.1. Mini-sum problems

A mini-sum FLP searches to locate a new facility that minimises the sum of the weighted distances between the new facility and the already existing facilities (data points or customers). They are often termed as "k-median facility location problems", "k" refers the number of facilities, $k \geq 1$. The new facility is nothing but the geometric median of the data points and no analytical solution is available if the number of data points is more than four.

The mini-sum location problem is expressed as follows:

$$\text{Minimise } f(x) = \sum_{i=1}^{m} w_i d(X_j, P_i), \tag{1}$$

where "X_j" is a new facility location after "j" iterations, "m" is the number of clients (data) points, "w_i" is the weight associated with the "ith" data point "P_i" and $d(X_j, P_i)$ is the distance between the "ith" data point "P_i" and "X_j".

The distance metric may be a rectilinear, Euclidean, great circle, ellipsoidal, or any other suitable one depends on the application.

Most of the authors consider the ED as the distance metric. A few researchers have used the rectilinear distance also in locating the facilities (Juel & Love, 1976; Aras et al., 2008).

When the data points are fairly separated, ED is not the appropriate metric and either GCD or ELD has to be used. The models proposed by Katz and Cooper (1980), Drezner and Wesolowsky (1978), and Xue (1994) consider earth as a perfect sphere. Not much work is available in FLP domain considering the earth as an ellipsoid.

2.2. Mini–max problems

In mini–max location problems, the objective is to minimise the maximum distance between the new facility and any existing facility. Both weighted and unweighted versions of such problems are possible. The primal application of this type is locating a public facility where the maximum distance between the final facility and farthest data point (customer) is minimised. It may also be referred to as a "k-centre facility location problem" in which "k" number of facilities is located, $k \geq 1$.

The objective function is:

$$\text{Minimise } f(x) = \text{maximum } \left[d(X_j, P_i)\right]; \quad \text{for } i = 1 \text{ to } m, \tag{2}$$

where "X_j" is a new facility location after "j" iterations and $d(X_j, P_i)$ is the distance between the "ith" data point "P_i" and "X_j". In this type also, a distance metric is used and again it may be a rectilinear, Euclidean, great circle, ellipsoidal, or any other suitable one depends on the application.

Exact algorithms are available to solve such problems in the Euclidean plane. Welzl (1991) algorithm is one such algorithm that solves for the minimum radius and centre in linear time. This can be generalised to compute the smallest enclosing balls or ellipsoids of point sets in higher dimensions. To the best of the author's knowledge, no algorithms are available in the literature for the minimum covering dome problems. Here, the data points and the facility are located on the curved exterior surface of the dome and the distance between any two data points is not Euclidean but, GCD.

The problem involves finding a minimum spherical dome that covers all the data points at its surface. If the data points are located on a flat surface then, this becomes a minimum covering circle problem. The four-point direction search algorithm could be used for this variant of FLP also but, iteratively.

3. Four-point direction search algorithm

This is a direction search heuristic algorithm that progresses in a direction that satisfies the specified criterion (objective function). This algorithm searches four points at a time instead of one. The four points are logically identified in the space. In its simplest form, each point is assumed to have two coordinates (x, y).

The algorithm is briefly explained in the following steps:

(i) The initial point is fixed usually based on some logical basis.
(ii) Four points are randomly selected in all four directions around the initial point by adding/subtracting a fraction (termed as "alpha") to the coordinates of the initial point.
(iii) All four points are checked for the objective function values.
(iv) The point that better satisfies the objective function among the four points is selected as the next point.
(v) Again, four points are randomly selected in all four directions around this new point by adding/subtracting a fraction to its coordinates.
(vi) The value of alpha is dynamically changed after a specific number of iterations.
(vii) The iteration is stopped once the terminating criterion is satisfied.
(viii) The alpha value is selected usually less than 0.2.

4. Methodology used

As discussed in Section 1, when the data points are widely separated on the earth's surface, locating an

optimal facility becomes difficult. The popular Weiszfeld algorithm fails to identify an optimal location in such a case. The four-point direction search algorithm could be effectively applied in such FLPs. The weighted distance metric can be any; ED, GCD, ELD, or driving distance, etc. It is assumed that we have the geodetic data (latitudes and longitudes) of all the data points.

The methodology can be better presented in the following steps:

(i) Convert the coordinates (latitudes and longitudes) of all data points to radians. ($180^\circ = \pi$ radians).
(ii) Finalize the weights, "w_i" associated with each data point P_i, $i = 1$ to m.
(iii) Compute the "Weighted Mass Centre" (arithmetic weighted mean of latitudes and longitudes)

$$\text{Mass centre, MC} = \frac{\sum_{i=1}^{m} w_i P_i}{\sum_{i=1}^{m} w_i}, \quad (3)$$

where "w_i" is the weight associated with the data point P_i and "m" is the number of data points.
(iv) Fix the value of the initial "alpha" (α) and the number of iterations. The value of alpha will be dynamically reduced after a fixed number of iterations, for example; after every 100 iterations.
(v) Locate four distinct points around the mass centre, MC separated by 90°. Following expression is used for this:
The coordinates of points, $CP_i = (MC \pm \alpha, MC \pm \alpha)$; $i = 1$ to 4.
(vi) Compute the sum of weighted distances, $\sum_{i=1}^{m} w_i d(X_j, P_i)$ to all the data points from these points.
(vii) Select the new point, NP with minimum total weighted distance as the next centre.
(viii) Randomly locate four distinct points around this new centre, NP using the following expression:
The new coordinates of points,

$$NCP_i = (NP \pm \alpha * (0.5 + 0.4 * \text{rand}\ ()), NP \pm \alpha * (0.5 + 0.4 * \text{rand}\ ())); i = 1\ \text{to}\ 4.$$

The random number generator generates a fraction between 0 and 1. Hence, the new points will be incremented with 0.5 times to 0.9 times the "α", in all four directions. This will also avoid getting any earlier points in the process.

(ix) Compute the sum of weighted distances to all the data points from these points (step vi).
(x) Select the point with minimum total weighted distance as the new centre.
(xi) If the terminating criterion is met, stop; else, go to step (viii).

Initially, a calibration procedure of parameter "alpha" was performed to obtain of good computational results.

5. Different distance metrics

The FLP problems essentially consider anyone distance metric. The distance metrics between two points generally used are rectilinear distance, ED, manhattan distance, minkowski distance, and hamming distance. ED is preferred for flat or two-dimensional surfaces. Hence, if the data points are spread over a smaller area on the earth's surface then only ED can be used that also will give an approximate value only.

$$\text{ED between points } (x1, y1) \text{ and } (x2, y2) = \sqrt{(x2-x1)^2 + (y2-y1)^2}. \quad (4)$$

However, when two points are considered on the earth's surface, two additional metrics are considered, GCD and ELD (Figure 2). If the earth is assumed to be a perfect sphere with a radius of 6371.009 km, GCD measures the shortest distance between two points on the earth's surface. GCD is based on spherical triangles whose sides form arcs of great circles (Das et al., 2001)

$$\text{GCD} = (\text{Earth Radius}) * (\text{Internal Spherical Angle in Radians}), \quad (5)$$

$$\text{Internal spherical angle between two points} = 2\sin^{-1}\sqrt{(sin^2 \Delta A + cos\ A1 * cos\ A2 * sin^2\ \Delta B)}, \quad (6)$$

where ΔA – half difference in latitudes of the two points $= (A1 + A2)/2$; $A1$ – latitude of the first point, radians; $A2$ – latitude of the second point, radians; ΔB – half difference in longitudes of the two points $= (B1 + B2)/2$; $B1$ – longitude of the first point, radians; $B2$ – longitude of the second point, radians.

This formula can be comfortably used for all points on the earth's surface except for antipodal points.

In reality, the earth is not a sphere but, an oblate spheroid. Earth is represented as an oblate spheroid since it is slightly flattened at the poles and bulges at the equator. An oblate spheroid is nothing but is an ellipsoid of revolution obtained by rotating an ellipse about its minor axis. The governing equation of an ellipsoid is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad (7)$$

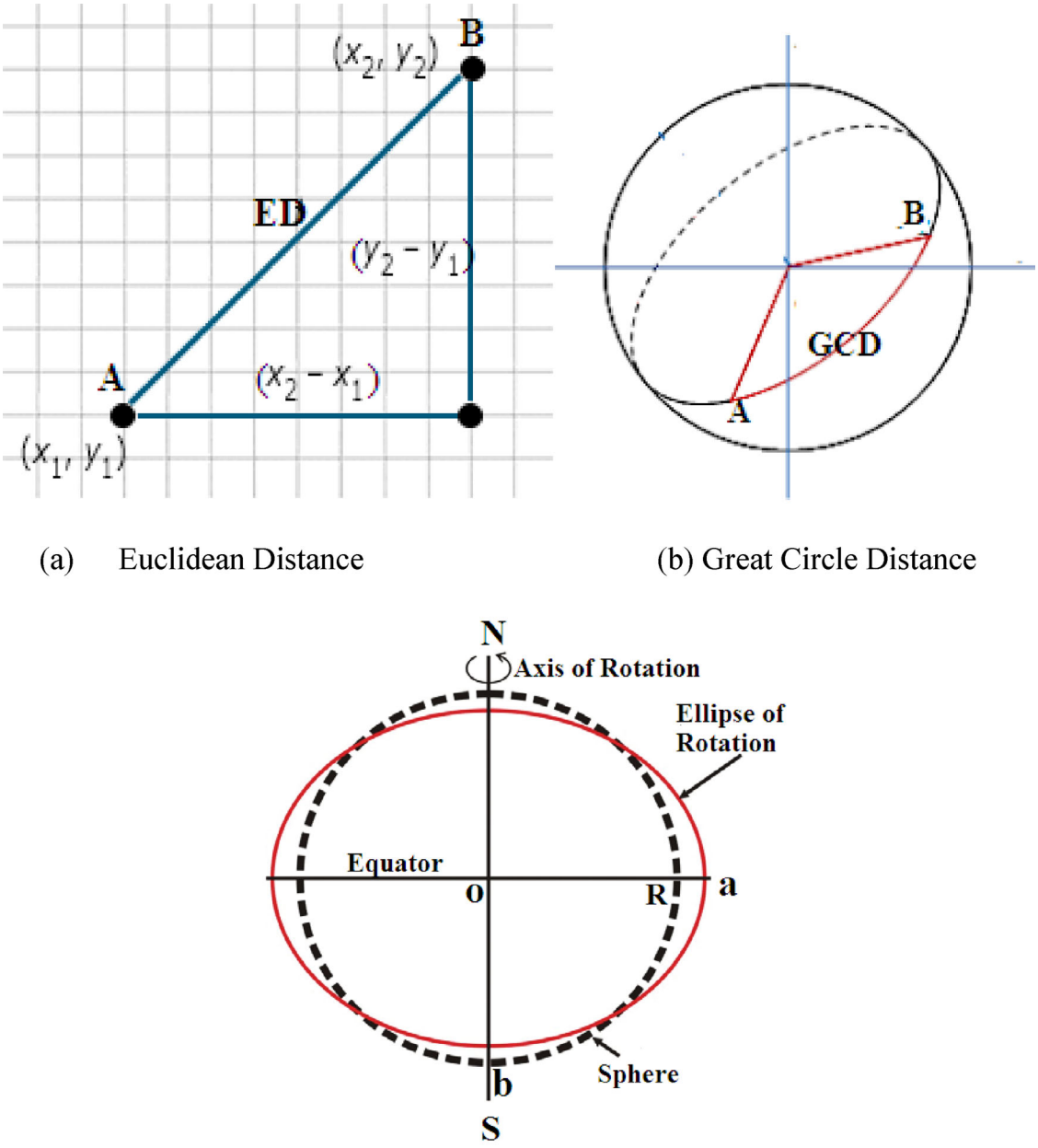


Figure 2. Distance metrics. (a) Euclidean distance. (b) Great circle distance. (c) Ellipsoidal model.

where x, y, and z are the Cartesian coordinates and a, b, and c are the semi-axes.

For a sphere, $a = b = c = R$ and the equation becomes,

$$x^2 + y^2 + z^2 = R^2. \tag{8}$$

For computing the ELD, Vincenty's formulae (Vincenty, 1975) are used which yield more accurate results than the GCD computed using the Haversine formula. The formulae were developed by Thaddeus Vincenty, a Polish-American geodesist who lived during the twentieth century. He considered the ellipsoidal model of the Earth (an oblate spheroid) and calculated the distance between a pair of latitude/longitude points.

In the ellipsoidal model:

Equatorial radius, $a = 6378.1370$ km;
Polar radius, $b = 6356.7523$ km and,
Mean Earth radius, $R = (2a + b)/3 = 6371.0088$ km.

Vincenty's formulae are two related iterative methods used to calculate the distance between two points on the surface of a spheroid. The first one is a direct method that computes the location of a point which is a given distance and azimuth (direction) from another point. The second one is an inverse method that computes the distance and azimuth between two given points. These Vincenty's formulae have been widely used because they are proved to be accurate to within 0.5 mm on the earth

Table 1. Lucia datasets.

Data point no.	Set A Locations	Set B Locations
1.	Mexico	Lamboano
2.	Oklahoma	Lhasa
3.	Cusco	Kuala Lumpur
4.	Algeria	Vorkuta
5.	Lusaka	Sudak
6.	Belgium	Marsa
7.	Sydney	Magadan
8.	Hobart	Cochabamba
9.	Seoul	Villa Puerto
10.	Kabul	Los Angeles
Mass centre (Radians)	0.173925, 0.435164	0.368514, 0.586099

ellipsoid (Daugherty, 2016). The formulae are not discussed in detail here as they involve complex expressions.

6. Validation of the algorithm

The formulae and methodology are coded in MATLAB 2012b and run in an i5 PC with 4 GB RAM. The codes are validated for the distance metrics by computing the distance between the Pune Airport (PNQ) with latitude and longitude, 18°34′55″N, 73°55′10″E (0.324316112, 1.290137687) radians and, Sydney Airport (SYD) with coordinates 33°56′45″S, 151°10′37″E (−0.592466559, 2.638535434) radians. The GCD obtained is 10036.25640 km between the two airports whereas, the ELD obtained is 10029.25345 km.

An online portal (Distance between Pune (PNQ) and Sydney (SYD), n.d.) reports the distances as 10036.260 and 10029.258 km respectively, and the actual distance as 10,029 km. The ELD reported by Vincenty's formulae is very close to the actual distance. The ED between these two airports is 9013.4 km which deviates significantly from the actual.

To validate the algorithm, two datasets available in the literature are considered (Cazabal-Valencia et al., 2016) and they are referred to as Lucia Set A and Lucia Set B (Table 1) in this paper. They used a "solver" to fix the facilities.

The facilities are located by running the four-point search algorithm (referred to as algorithm AB). The maximum number of iterations is limited to 1000 and in each case, 10 trials are conducted. Not much difference is observed in the results between the trials. The mass centre is taken as the initial centre in all the cases.

For Lucia dataset A; total GCD varies from 85679.54813 to 85679.58605 km (a variation of 0.03792 km) and the ELD varies from 85603.32635 to 85603.35251 km (a variation of 0.02616 km). For Lucia dataset B; the total GCD obtained is 70654.19805 km and ELD obtained is 70588.50724 km in all 10 trials.

The comparison is listed in Table 2.

Instead of taking the mass centre as the initial centre, two more options are tried. In the first case, (0, 0) is taken as the initial centre, and in the second case; the combination of (extreme value of latitude, extreme value of longitude) in the given dataset is the initial point. In both cases, the results obtained are almost the same.

Since the data points are widely separated; the GCD between the mass centre and final centre for Lucia dataset A is 5535.0 km and, the GCD between the mass centre and final centre for Lucia dataset B is 4015.9 km.

7. Results and discussion

After validation; two problems are considered, the first one is a random one and the other one is a real-time problem considering the 661 districts of India (2011 census). The objective is to locate a single facility to minimise the total distance in Case (i) and minimise the maximum radius of a covering spherical dome in Case (ii).

7.1. Case I: single facility location

7.1.1. A central facility for a random dataset

In line with the Lucia datasets, 10 data points are randomly selected covering all continents. The latitudes, longitudes, and the mass centre of 10 random data points are presented in Table 3. Using the four-point search algorithm 'AB', the facilities are located and distances are computed (Table 4).

The initial alpha is fixed at 0.0512 and the mass centre is taken as the initial centre. After every 100 iterations; alpha gets halved and the number of iterations is limited to 1000. Ten trials are conducted and in all 10 trials, we get the GCD as 67835.95306 km and the facility is at (1.1745, 1.0884) radians. Similarly, the total ELD reported is 67755.91956 km and the centre is at (1.1764, 1.0861) radians. For this dataset, the ED is reported as 59664.00701 km. GCD is the maximum and ED is the minimum. The difference between GCD and ELD is 80.034 km whereas; the difference between GCD and ED is 8171.9 km.

When the iterative Weiszfeld algorithm is used to fix the facility, the location is at (0.60017, 0.79761) radians and the total GCD is 70798.34652 km which is 2962.4 km more than the GCD obtained using the four-point search algorithm. The facilities located using GCD and ELD are different and separated by a GCD of 13.356 km or ELD of 13.383 km. Though the difference in the total distance reported for ED is high from GCD (less by 8171.9 km), the facilities are closely located, the distance being 92.325 km. Whereas, the facilities located by GCD (AB) and GCD (Weiszfeld) are separated by 3754.5 km. This

Table 2. Lucia datasets – comparison of results using four-point search algorithm "AB".

S. no.	Distance metric	Dataset A Facility (latitude, longitude)	Total distance, km	Dataset B Facility (latitude, longitude)	Total distance, km
1.	GCD (Lucia)	0.88135, 0.058565	85679.42703	1.1010, 1.0780	70654.08890
2.	ELD (Lucia)	0.88135, 0.058571	85610.24031	1.0880, 1.0696	70636.37066
3.	GCD (AB)	0.881348, 0.058576	85679.54813	1.1010, 1.0780	70654.19805
4.	ELD (AB)	0.881351, 0.058575	85603.32635	1.1164, 1.0876	70588.50724

Table 3. Random dataset.

S. no.	Location	Latitude, dec.	Longitude, dec.
1.	Yukon, Canada	60.170638	−130.827364
2.	Kamchatka Krai, Russia	62.424437	169.684973
3.	Durazno, Uruguay	−33.195543	−55.429532
4.	London, UK	51.5085300	−0.1257400
5.	Ihosy, Madagascar	−22.488918	45.657826
6.	Hamrin Mountain, Iraq	35.050944	43.636343
7.	Nenets Autonomous Okrug, Russia	68.031820	61.372730
8.	Thellai, India	12.776006	79.028060
9.	Omakau, New Zealand	−45.062944	169.629765
10.	Hulunbuir, Inner Mongolia, China	49.753488	124.590197
Mass centre	Al Udayd, Saudi Arabia	23.8968458	50.7217258

Table 4. Random dataset – four-point search algorithm "AB".

S. no.	Distance metric	Facility latitude, rad.	Facility longitude, rad.	Total distance, km
1.	GCD (AB)	1.1745	1.0884	67835.95306
2.	ELD (AB)	1.1764	1.0861	67755.91956
3.	ED (AB)	1.1874	1.0712	59664.00701
4.	GCD (Weiszfeld)	0.60017	0.79761	70798. 34645

Table 5. 661 districts of India (2011 Census) – four-point search algorithm "AB".

S. no.	Distance metric	Facility latitude, rad	Facility longitude, rad.	Total distance, km
1.	GCD (AB)	0.41766	1.41085	537412.97730
2.	ELD (AB)	0.41763	1.41076	536163.90748
3.	ED (AB)	0.41770	1.41072	535530.00206
4.	GCD (Weiszfeld)	0.41513	1.41118	537489.58801

endorses the efficiency of the four-point search algorithm 'AB'.

Since the data points are widely separated, the distance between the mass centre and the final facility is high. They are separated by a GCD of 4892.5 km or by an ELD of 4895.3 km.

7.1.2. A central facility for 661 Indian districts

In this case, the problem is to fix a single facility to serve the 661 district headquarters of India (2011 census). All data points are assumed to have equal weights and hence the distance is the base metric for fixing the facility. In this work, only a single facility is estimated without assigning any weight to the data points and the results are presented in Table 5. In the 10 trials conducted, total GCD varies from 537412.97730 to 537412.97944 km, a difference of 0.00214 km, and the facility is at (0.41766, 1.41085) radians corresponding to the minimum distance. Similarly, ELD is estimated using Vincenty's formulae and the total distance is reported as 536163.90748 km and the facility is located at (0.41763, 1.41076) radians. The ELD varies from 536163.90748 to 536163.90779 km, the difference being 0.00031 km. For the same dataset, ED varies from 535530.00206 to 535530.0063 km (a difference of 0.00424 km).

When the iterative Weiszfeld algorithm is used to fix the facility, the location is at (0.41513, 1.41118) radians and the total GCD is 537489.58801 km which is 76.6107 km more than the GCD obtained using the four-point search algorithm.

The GCD between the mass centre and facility location is 79.897 km (i.e. they are nearer to each other). The facilities computed using any method and distance metric are closely located. Hence, the facility located using ELD is mapped and shown in Figure 3 for reference. The place is Ubara located in the state of Madhya Pradesh.

It is to recall that for the widely separated Lucia datasets A, B, and the new random dataset, the distance between the mass centre and the final location is high.

7.2. Case II: application to the minimum covering spherical dome problem

The algorithm requires only a minor modification. In steps (vi) and (ix) of the "Methodology"; instead of the sum of the distances, the maximum distance between the temporary facility and any data point is



Figure 3. Facility based on ELD (Ubara, Madhya Pradesh).

Table 6. Radius of minimum covering spherical dome.

Facility latitude, rad	Facility longitude, rad.	Minimum great circle radius, km	Total GCD, km
Random dataset of 10 points			
0.13269	0.97653	12455.35575	78838.91365
661 Districts of India			
0.37757	1.45909	1698.1	581223.91626
657 Districts of India (excluding the four districts located in islands)			
0.38144	1.43353	1602.5	552235.21654

estimated. The final location is a point that minimises the maximum distance.

The algorithm is applied to the random set of 10 points and 661 districts of India (2011 census) the results of which are presented in Table 6. Only GCD is considered for the analysis.

Here, the objective is to cover all the data points with a dome having a minimum radius. The total

distance is ignored. Hence, the results show (Table 6) high total GCDs, 78838.91365 km for the random dataset and, 581223.91626 km for the Indian districts' case. If the four-island districts of India are ignored, the total GCD comes down to 552235.21654 km. In such a case, the minimum radius is 1602.5 km which is 95.6 km less.

Similarly, if ED is used in the place of GCD, the results for the planar minimum covering circle are obtained.

8. Conclusion and future work

The proposed algorithm is validated using the datasets available in the literature for GCD and ELD. Also, the model is demonstrated for locating a central facility considering 10 randomly generated points spread across the continents and 661 districts headquarters of India (2011 census). Equal weights are considered for all the data points and only one cluster is considered to locate a facility in both the FLPs analysed. In all the cases, the algorithm converges to a global minimum resulting in optimal/near-optimal facilities considering the total distance and minimum covering spherical dome. Since the algorithm analyses four points at a time, the computation time is high which can be optimized by changing the value of alpha and the number of iterations. The main strength of the proposed algorithm is that it can be applied to optimally locate the facility on a plane, sphere and ellipsoid surfaces without modifying it; using the appropriate distance metric only need to be ensured. If the data points are closely located then, the mass centre and the final facility may be close to each other. In such a case, we can select smaller alpha with a reduced number of iterations. Future work includes multi-facility locations assuming different weights for each data point.

Acknowledgements

The authors are sincerely thankful to the anonymous referees, who provide constructive comments to improve the technical content and presentation of the paper. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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